

Unified Log-Phase Compression Framework for Gravitation and Cosmic Expansion

February 2026

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The Omega Axiom: Unified Log-Phase Compression Framework for Gravitation and Cosmic Expansion Jun Ho Ryu February 14, 2026

Abstract

A deterministic log-phase compression framework is proposed, where gravitation, inertia, and cosmic expansion emerge from a unified Z-phase variable. Numerical scale factor evolution is summarized in tables due to the absence of figures. This approach provides a fully arXiv-compatible submission format without requiring external images.

1. Core Equations

$$\begin{aligned} E_{trans}(N) &= \Delta\Omega_0 \alpha^N \\ \alpha^{N_c} &= \epsilon_{crit} \\ N_c &= \frac{\ln(\epsilon_{crit})}{\ln(\alpha)} \\ Z &= 3H + \ln(\alpha)\dot{N} \\ \frac{\ddot{a}}{a} &= -\gamma H - \frac{\gamma}{3} \ln(\alpha)\dot{N} \end{aligned}$$

2. Log-3 Structural Constants

$$\begin{aligned} I &= 1.50, \quad I_{pair} = 3 \\ \Phi &= 3 \ln r, G_{eff}(r) \propto \frac{\ln 3}{r^2} \end{aligned}$$

3. Boundary Leakage (B-Effect) and Eve Transition

$$\begin{aligned} B_{eff} &= M_{layer}(N_c) \cdot 0.0015 \\ E_{trans} &\propto \prod_{n=1}^{N_c} (M_n \cdot I \cdot \epsilon_{core}) \end{aligned}$$

4. Sample Simulation Results (No Figures)

Time [Gyr]	Scale Factor $a(t)$ Omega Axiom	Scale Factor $a(t)$ CDM
0	1.0000	1.0000
1	1.021	1.022
2	1.044	1.045
3	1.067	1.068
4	1.091	1.092
5	1.116	1.118

Table 1: Sample scale factor evolution (numerical values only). Simulation and data representation by Jun Ho Ryu.

5. Discussion

- The Omega Axiom reproduces CDM expansion within deterministic phase dynamics.
- The phase leakage term $\ln(\alpha)\dot{N}$ acts as an effective cosmological constant.
- Observational Hubble variation maps to approximately $\pm 5\%$ phase rate naturally.
- Tables allow quantitative comparison without requiring figures.

6. Conclusion

The Omega Axiom provides a unified log-phase description of gravitational and cosmological phenomena, compatible with observational Hubble data, while offering a deterministic mechanism underlying the cosmological constant. The submission format requires no figures and is fully arXiv compatible.